

RBC X BCCAI X SFU Beedie Agribusiness Analytics Hackathon 2026¹²

“A country that **cannot feed itself**, fuel itself or defend itself has few options,” said Prime Minister Mark Carney at 2026 Davos Summit.

Food security has emerged as a pressing national priority, one that RBC and BCCAI recognized well before it entered mainstream policy debate. Independent farmers face a structural challenge that limits their resilience: the inability to accurately estimate and project total operating costs. This gap constrains financial planning, weakens their position in contract negotiations with buyers, and limits their access to capital.

The convergence of rising input costs, volatile energy prices, and increasing climate variability has made this challenge more acute. Precision agriculture technologies offer a path forward, enabling farmers to move from rules of thumb to data-driven decisions, but the upfront capital required is a barrier that many independent operators cannot overcome on their own.

BCCAI and RBC have been brainstorming and exploring a concept that would enable independent farmers in British Columbia to improve their production efficiency and cost effectiveness by using precision agriculture equipment and technologies. The initiative draws on the framework established by the USDA Precision Agriculture Loan Program³, a bill introduced in the U.S. House and Senate in 2024, and adapts it proactively for the Canadian context, tailored to B.C.'s specific agricultural profile and regulatory environment. Rather than waiting for U.S. legislative outcomes, this concept potentially positions RBC and BCCAI as architects of a made-in-Canada solution. As part of this concept, wraparound support services, including research assistance, application writing guidance, and technical advisory services, could help farmers develop strong, production, competitive farming operations that are resilient.

The Central Question

The key feasibility consideration is whether precision technology can meaningfully improve independent farmers' financial decision-making, and if so, how that value can be demonstrated.

You are thus asked to use the data described below for a hypothetical Controlled Environment Agriculture (CEA) operator (commonly referred to as a greenhouse) in B.C., GreenLeaf CEA, along with **any supplementary external data or industry benchmarks**, to build dashboards that link the metrics

¹ This case was prepared by Michael Johnson and Jason Ho, Beedie School of Business, Simon Fraser University. The information presented in this case study has been simplified, adapted, or fictionalized for educational, teaching, and learning purposes

² We extend our gratitude to Melanie Lantz, Lorna McKercher and the agriculture team at RBC, Rahul Singh of BCCAI, Tony Kalf, Flora Yu and the team at Windset, and everyone else who assisted, though the responsibility for any mistakes lies solely with us.

³ <https://www.nifa.usda.gov/grants/programs/precision-geospatial-sensor-technologies-programs/adoption-precision-agriculture>

enabled by sensors and other precision technologies to yields, enabling ROI and/or other financial impact calculations of various operational decisions.

GreenLeaf CEA

GreenLeaf Controlled Environment Agriculture (CEA) operates eight farm locations across British Columbia, supported by twenty greenhouse units and one hundred twenty experimental microplots. Together, these sites produce premium tomatoes, peppers, cucumbers, and strawberries for local grocers and restaurants under a mix of greenhouse and mixed production systems in different climate zones. The company has invested heavily in infrastructure and sensors but still faces intense economic pressure from rising input costs, volatile energy prices, and labor constraints.

Despite operating in controlled environments, GreenLeaf's managers live with constant uncertainty. Input costs can swing week to week, microclimates still create heat and disease pressure, and yield and quality are only known with certainty at the end of the season. Historically, many input decisions have been made based on rules of thumb - such as "high nitrogen" for more yield - without clear evidence of the true impact on plant stress, quality, or profit. The result is a risk of overspending on inputs that fail to pay back.

To tackle this, GreenLeaf has designed an on-farm experimentation program across one hundred twenty microplots. Each plot is assigned to a specific combination of crop, fertilizer or pest strategy (for example, Control, Low Nitrogen, High Nitrogen, Integrated Pest, Reduced Pest, High Light, or Shade), plant density, and greenhouse infrastructure. Over a season, the company records daily environmental and crop stress signals, targeted input applications, scouting observations, market input price conditions, and end of season financial outcomes for every plot.

Their precision technology data now goes beyond simple monitoring. The daily sensor data now includes alert flags, recommended actions, whether an action was taken, action delay, and short-run post-action stress change, while the applications table identifies whether an intervention was a precision action versus routine management. This means the dashboard can now show not only what conditions existed, but also whether GreenLeaf responded, how quickly it responded, and whether those responses were associated with better outcomes later in the season.

Your mission: transform GreenLeaf's experimental, agronomic, and financial data into a dashboard that helps decision-makers see which plots, treatments, and interventions delivered the best in-season response and end-of-season profitability. The final dashboard should bridge environmental monitoring, management action, and business return so GreenLeaf can make faster and smarter operating decisions.

You will present your dashboard as a demonstration to B.C. farmers who are considering investing in technology to help them improve their operation. Your presentation should make the value of precision technology concrete and legible to a non-technical audience. Several use cases you may want to use to guide your dashboard development and presentation:

(A) Weekly alert triage – “Where should I act first this morning?”

In a given week of the season, the greenhouse sensor system flags a spike in high-stress events across many plots. The operations manager has limited labor on Monday morning and must decide which plots the crew should visit first and where precision (alert-driven) actions are most justified. The dashboard should show, for a selected week, how many high-stress events occurred, how quickly the team has been responding, and how often alerts led to actions. It should provide a ranked list of “plots needing attention” plus a simple drill-down timeline for any plot, showing stress, costs, and action days. Using this, the

manager can focus scarce resources on the plots where fast, precision interventions are most likely to reduce stress, improve yield and maintain finances.

(B) Seasonal evaluation of precision actions – “Is this system worth financing?”

Consider the following scenario: a farm owner is preparing for a meeting with RBC to discuss financing for continuing or expanding their precision agriculture system. They want to know whether using alerts, sensors, and precision actions has clearly improved plant health, operational and financial performance compared with relying only on routine practices. The dashboard should summarize, over the whole season, how often plants were in high stress, how quickly and reliably the team responded, and how much was spent on precision versus routine actions. It should highlight periods or plots where high precision spending produced visible stress reduction versus periods where costs were high and stress stayed high, signaling inefficiency. Using these patterns, the farmer can tell a simple, evidence-based story to the lender about how precision technology helps manage risk, protect yield, and justify additional investment.

(C) Open scenario – “Design your own decision story”

Teams may also define a third user case of their own, grounded in the provided data and farming context. For example, they might focus on comparing crops or treatments, evaluating performance of specific greenhouses, or identifying “at-risk” plots before a heatwave. The dashboard should clearly state the question, show which metrics and visuals answer it, and explain why this scenario matters to a farmer or lender. This open scenario allows teams to demonstrate creativity while still building on the same stress, alerts, action, and cost data as the case provides.

GREENLEAF DATASET DETAILS (9 datasets in this [OneDrive folder](#))

1. FARMS (8 rows – farm-level context)

- **farm_id** (Dimension, key): Unique identifier for each farm location. Values look like short codes (e.g., “F01”, “F02”) and are used to join to other tables.
- **farm_name** (Dimension, label): Human-readable name of each farm (e.g., “BC Harvest 1”). This is what you show on labels, tooltips, and filters.
- **region** (Dimension, geography): Municipality or region for the farm (e.g., Abbotsford, Maple Ridge). Use for regional breakdowns or maps.
- **primary_crop** (Dimension, category): Main crop associated with the farm (e.g., Tomato, Pepper, Strawberry). Useful for high-level crop mix views.
- **greenhouse_acres** (Measure, numeric): Total greenhouse area for the farm in acres. Use for scale normalization (e.g., revenue per greenhouse acre).
- **production_system** (Dimension, category): Production model used at the farm (e.g., Greenhouse, Mixed). Good for comparing full-greenhouse vs mixed systems.
- **climate_zone** (Dimension, category): Climate classification for the farm (e.g., Coastal, Interior, Fraser Valley). Use to study how climate affects performance.

2. GREENHOUSES (20 rows – greenhouse-unit context)

- **greenhouse_id** (Dimension, key): Unique identifier for each greenhouse (e.g., “GH001”). Primary join key from daily, input, and scouting tables to greenhouse details.

- **farm_id** (Dimension, foreign key): Links each greenhouse back to its farm (joins to `farms.farm_id`). Enables roll-up from greenhouse to farm.
- **crop** (Dimension, category): Primary crop grown in that greenhouse (e.g., Tomato, Strawberry, Cucumber, Pepper). Note this is greenhouse-level, not plot-level.
- **structure_type** (Dimension, category): Greenhouse structure material (e.g., Glass, Polyfilm). Use for performance comparisons between infrastructure types.
- **heating_system** (Dimension, category): Type of heating system in the greenhouse (e.g., Hydronic, Natural Gas, Heat Pump). Use for cost or efficiency analysis.
- **irrigation_system** (Dimension, category): Irrigation technology used (e.g., Drip, EBB-Flow, Fertigation). Link to water use and stress outcomes.
- **capacity_plants** (Measure, numeric, count): Approximate maximum plant capacity in that greenhouse. Can normalize yield or revenue per plant.
- **sensor_density_per_100m2** (Measure, numeric): Number of sensors per 100 square metres of growing area. Higher values mean more granular monitoring.

3. PLOTS (120 rows – experimental microplots)

- **plot_id** (Dimension, key): Unique identifier for each microplot (e.g., “P0001”). Central identifier for joining to daily, inputs, scouting, and season outcomes.
- **greenhouse_id** (Dimension, foreign key): Greenhouse that contains the plot (joins to `greenhouses.greenhouse_id`). Lets you see plots within each house.
- **farm_id** (Dimension, foreign key): Farm that owns the plot (joins to `farms.farm_id`). Useful for farm-level aggregation and filtering.
- **crop** (Dimension, category): Crop grown in that plot (e.g., Strawberry, Tomato, Pepper, Cucumber). Plot-level crop can differ from greenhouse default if designed that way.
- **treatment_block** (Dimension, experimental factor): Experimental block such as A/B/C/D used in randomized designs to reduce bias. Use in analysis of treatment effects.
- **treatment** (Dimension, treatment label): The primary experimental strategy applied to the microplot. This is a single categorical dimension where the seven levels (Control, Low N, High N, Integrated Pest, Reduced Pest, High Light, Shade) are mutually exclusive. Each plot is assigned exactly one label, representing a unified operational strategy rather than a combination of multiple factors. The "Control" serves as the historical baseline (routine management) against which the "Precision Benefit" of other alert-driven interventions is measured.
- **plot_area_m2** (Measure, numeric): Physical size of the plot in square metres. Use for per-area normalization (e.g., yield kg/m², cost per m²).
- **plant_density_plants_m2** (Measure, numeric): Average number of plants per square metre (stems/m²). Use to examine how density influences yield and stress.

- `row_orientation` (Dimension, category): Direction of rows (North, South, East, West). This affects light exposure and can be used in exploratory analysis.

4. **DAILY_SENSOR_READINGS** (25,200 rows – daily plot-level signals and alerts)

- `date` (Date): Calendar date for the observation. Combine with `plot_id` to identify one row per plot per day.
- `plot_id` (Dimension, foreign key): Plot measured that day (joins to `plots.plot_id`).
- `greenhouse_id` (Dimension, foreign key): Greenhouse where the plot resides (joins to `greenhouses.greenhouse_id`).
- `farm_id` (Dimension, foreign key): Farm associated with the plot/greenhouse (joins to `farms.farm_id`).
- `crop` (Dimension, category): Crop in that plot on that day. Duplicates context from `plots` for convenience.

Environmental and substrate measurements (all Measures, numeric):

- `air_temp_c`: Air temperature in the greenhouse in degrees Celsius.
- `relative_humidity_pct`: Relative humidity in percent (0–100).
- `vpd_kpa`: Vapor pressure deficit in kilopascals. Important driver of transpiration and stress.
- `co2_ppm`: CO₂ concentration in parts per million. Higher values usually mean more aggressive CO₂ enrichment.
- `light_ppfd`: Photosynthetic photon flux density (PPFD), typically measured in $\mu\text{mol}/\text{m}^2/\text{s}$, representing light available for photosynthesis.
- `substrate_moisture`: Fractional moisture content of the root-zone substrate (0–1 scale). Values outside the target band can signal over- or under-watering.
- `substrate_ec`: Electrical conductivity of the substrate solution in mS/cm, capturing nutrient strength and salt load.
- `substrate_ph`: pH of the substrate solution, indicating acidity/alkalinity of the root environment.
- `canopy_temp_c`: Leaf or canopy surface temperature in degrees Celsius, which can diverge from air temperature under stress.

Risk and stress indices (all Measures, numeric 0–1):

- `pest_pressure_index`: Modeled pest pressure on a 0–1 scale (0 = none, 1 = extreme risk).
- `disease_risk_index`: Modeled disease risk on a 0–1 scale (0 = none, 1 = extreme risk).
- `plant_stress_index`: Composite plant stress score on a 0–1 scale (0 = healthy, 1 = critical stress).

Alert and response fields:

- `alert_type` (Dimension, category): Type of alert triggered by sensor conditions, such as High VPD or Low Moisture. When no alert fires, this may be blank or null.

- **alert_flag** (Indicator, 0/1): Numeric flag indicating whether an alert was active on that plot and date (1 = alert, 0 = no alert). Use for counts or percentages of alert days.
- **recommended_action** (Dimension, category): The suggested action associated with the alert (e.g., Increase Irrigation, Increase Venting). This is prescriptive guidance, not necessarily what was done.
- **action_taken** (Indicator, 0/1): Numeric flag indicating whether the recommended (or a related) action was actually taken (1 = action taken, 0 = no action).
- **action_delay_days** (Measure, numeric): Number of days between the alert and the action applied. Zero means same-day response; larger values mean delayed response.
- **post_action_stress_delta_3d** (Measure, numeric): Change in **plant_stress_index** over the three days after the action window. Negative values often indicate improved stress, positive values indicate worsening.

5. DAILY_INPUT_COSTS (25,200 rows – daily plot-level input costs and actions)

Note: This table provides a more granular view of input costs on selected days during the season, broken out by routine and precision actions. It is designed to support exploration of cost patterns and illustrative examples, but it is not a full, line-by-line reconstruction of the accounting totals reported in **season_summary** table.

- **date** (Date): Calendar date for the record. Combine with **plot_id** to identify one row per plot per day.
- **plot_id** (Dimension, foreign key): Plot where costs and actions are recorded for that day (joins to **plots.plot_id**).
- **greenhouse_id** (Dimension, foreign key): Greenhouse that contains the plot (joins to **greenhouses.greenhouse_id**).
- **farm_id** (Dimension, foreign key): Farm associated with the greenhouse/plot (joins to **farms.farm_id**).
- **daily_energy_cost** (Measure, currency): Total energy cost (e.g., heating, lighting, ventilation) allocated to this plot on that date.
- **daily_fertilizer_cost** (Measure, currency): Fertilizer input cost allocated to this plot on that date.
- **daily_labor_cost** (Measure, currency): Labor cost associated with work on this plot on that date (both precision and routine activities).
- **daily_pesticide_cost** (Measure, currency): Pesticide or crop-protection cost for this plot on that date.
- **daily_water_cost** (Measure, currency): Irrigation water cost for this plot on that date.
- **daily_total_input_cost** (Measure, currency): Total input cost for this plot on that date, equal to energy + fertilizer + labor + pesticide + water.

- `daily_precision_cost` (Measure, currency): Portion of `daily_total_input_cost` associated with precision (sensor-/alert-driven) actions on this plot.
- `daily_precision_actions_count` (Measure, count): Number of precision actions carried out on this plot on that date.
- `daily_total_actions_count` (Measure, count): Total number of actions (precision + routine) carried out on this plot on that date.
- `daily_routine_cost` (Measure, currency): Portion of `daily_total_input_cost` associated with routine (scheduled, baseline) actions on this plot.

6. INPUT_APPLICATIONS (4,613 rows – plot-level management actions and costs)

- `date` (Date): Date when the input was applied or the work event occurred. Multiple applications can share the same date and plot.
- `plot_id` (Dimension, foreign key): Plot receiving the input (joins to `plots.plot_id`).
- `greenhouse_id` (Dimension, foreign key): Greenhouse where the plot is located.
- `farm_id` (Dimension, foreign key): Farm where the plot resides.

Application details:

- `input_type` (Dimension, category): Type of input or resource applied. Typical values include Fertilizer, Pesticide, Water, Labor, Energy.
- `quantity` (Measure, numeric): Physical quantity of the input applied (e.g., kg of fertilizer, litres of water, labour hours, kWh of energy). The units depend on `input_type`.

Cost metrics (all Measures, numeric, currency):

- `unit_cost`: Cost per unit of the input in CAD (e.g., CAD/kg, CAD/L, CAD per hour).
- `total_cost`: Cost of this application event in CAD, typically $\text{quantity} \times \text{unit_cost}$.

Management intent and precision labeling:

- `reason` (Dimension, category): Operational trigger for the application (for example, Shock response, Growth phase, Pest response, Irrigation adjustment, Routine). It describes “why now?”.
- `purpose` (Dimension, category): Higher-level purpose classification, such as Routine versus Response to alert. Use to separate background management from targeted interventions.
- `is_precision_action` (Indicator, 0/1): Numeric flag indicating whether this particular application is considered a precision action (1) or not (0). Use for counting precision interventions and comparing cost/outcome.

7. SCOUTING_OBSERVATIONS (1,800 rows – human agronomist observations)

- `date` (Date): Date of the scouting visit or inspection. Typically at a coarser frequency than daily sensors (e.g., weekly).
- `plot_id` (Dimension, foreign key): Plot inspected (joins to `plots.plot_id`).

- `greenhouse_id` (Dimension, foreign key): Greenhouse containing the plot.
- `farm_id` (Dimension, foreign key): Farm containing the greenhouse.

Observed plant condition:

- `blossom_count` (Measure, numeric): Number of blossoms per plant recorded during scouting. Higher values often correlate with yield potential.
- `pest_severity` (Measure, numeric 0–1): Visual rating of pest damage or pressure from 0 (none) to 1 (severe).
- `disease_severity` (Measure, numeric 0–1): Visual rating of disease symptoms from 0 (none) to 1 (severe).
- `leaf_color` (Dimension, category): Categorical description of foliage color, typically Excellent, Good, Fair, or Poor, used as a quick health proxy.
- `notes_flag` (Dimension, category): Coded summary of notes highlighting issue severity (None, Minor, Moderate, Severe). Helps flag plots needing follow-up.

8. MARKET_AND_INPUT_PRICES (840 rows – external price context)

- `date` (Date): Date associated with the price observation, typically a week-ending date.
- `item` (Dimension, category): Type of input priced (Fertilizer, Pesticide, Energy, Water).
- `price` (Measure, numeric, currency): Observed unit price for that item in CAD.
- `unit` (Dimension, category): Unit associated with the price (e.g., CAD/kg, CAD/L, CAD/kWh). Use together with quantity fields in `input_applications` for external benchmarking.

9. SEASON_SUMMARY (120 rows – plot-level end-of-season results)

Note: This table summarizes each plot's overall financial performance based on the farm's accounting system for the full season. It is the primary source for total revenue, total cost, profit, ROI, and the estimated profit contribution from precision practices at the plot level.

- `plot_id` (Dimension, key/foreign key): Plot identifier (joins to `plots.plot_id`). This table holds one row per plot for the season.

Production and quality outcomes (all Measures, numeric):

- `season_yield_kg_m2`: Total seasonal yield per square metre of plot area, in kilograms per m².
- `marketable_ratio`: Fraction of yield that is marketable (0–1), where 1 means all produce is premium grade.

Financial performance (all Measures, numeric, currency or ratio):

- `season_revenue_cad`: Total seasonal revenue from that plot in CAD. Usually derived from yield and price assumptions.
- `total_cost_cad`: Total seasonal cost allocated to that plot in CAD, including inputs and fixed allocations.

- `season_profit_cad`: Seasonal profit in CAD, generally `season_revenue_cad` minus `total_cost_cad`.
- `season_roi`: Seasonal return on investment, calculated as $(\text{revenue} - \text{cost}) / \text{cost}$. Values above 1 indicate “>100% return” relative to cost.
- `precision_benefit_cad`: Estimated benefit in CAD attributable to precision management (e.g., targeted responses vs routine). Use to highlight where precision actions paid off financially.

Appendix: Glossary for the GreenLeaf CEA Case

This glossary explains the business, agronomic, and data terms used across the GreenLeaf case.

Business and operating context

- **Controlled Environment Agriculture (CEA):** Food production in managed growing environments where temperature, humidity, irrigation, light, and related conditions are actively controlled rather than left entirely to outdoor weather.
- **Farm:** A top-level operating location in the dataset. Each farm has a unique farmid, a name, a region, a climate zone, a production system, and greenhouse acreage.
- **Greenhouse:** A production unit within a farm, identified by greenhouseid. Greenhouses are nested inside farms and have their own crop, structure, heating system, irrigation system, capacity, and sensor density.
- **Plot / Microplot:** The core experimental unit in the case, identified by plotid. A plot is a small growing area inside a greenhouse where a specific treatment combination is applied and tracked over time.
- **Experimental program:** The on-farm testing approach GreenLeaf uses to compare different treatments, management styles, and growing conditions across many plots during the season.
- **Production system:** The farm-level production model, such as Greenhouse or Mixed, used to compare operating contexts across locations.
- **Climate zone:** A broad environmental classification for each farm, such as Coastal, Interior, or Fraser Valley, used to study performance differences by regional growing conditions.

Experimental design terms

- **Treatment:** The primary experimental strategy applied to a microplot. In this case, treatment is a single categorical dimension with seven mutually exclusive levels: Control, Low N, High N, Integrated Pest, Reduced Pest, High Light, and Shade. Each plot receives exactly one treatment, representing a unified management decision rather than a multi-factor combination.
- **Control:** A baseline treatment with no special experimental change, used as the reference point for comparing the effect of other treatments.
- **Low N / High N:** Fertilizer treatments that apply lower or higher nitrogen levels than the baseline. These are designed to test whether more or less nitrogen improves yield, quality, stress, or profit.
- **Integrated Pest:** A treatment strategy that combines pest management approaches in a more targeted or balanced way than a simple standard spray approach.
- **Reduced Pest:** A treatment strategy with less pest-control input than a standard approach, useful for testing whether lower intervention can still protect crop outcomes.
- **High Light / Shade:** Treatments that change the amount of light reaching the crop, helping students analyze how light conditions affect stress, quality, and profitability.

- **Treatment block:** A grouping variable such as A, B, C, or D used in a randomized experimental design to reduce bias and make treatment comparisons fairer.
- **Plant density:** The number of plants per square metre (plantdensityplantsm2) in a plot. Higher density can raise output per area but may also increase competition and stress.
- **Row orientation:** The direction of plot rows, such as North, South, East, or West, which may affect light exposure and microclimate conditions.

Sensor and environment terms

- **Daily sensor reading:** One plot-day observation in the daily_sensor_readings table. Each row captures environmental conditions, modeled risks, and alert-response fields for one plot on one date.
- **Air temperature (airtempc):** The measured air temperature in the greenhouse, in degrees Celsius.
- **Relative humidity (relativehumiditypct):** Moisture in the air expressed as a percent from 0 to 100.
- **VPD / Vapor pressure deficit (vpdkpa):** A measure of the drying power of the air, in kilopascals. In practice, higher VPD often means plants lose water faster and may become stressed if irrigation and climate control do not keep up.
- **CO2 (co2ppm):** Carbon dioxide concentration in parts per million. Higher values can indicate more aggressive CO2 enrichment to support plant growth.
- **PPFD / Light (lightppfd):** Photosynthetic photon flux density, a measure of light available for photosynthesis.
- **Substrate moisture (substratemoisture):** Root-zone moisture on a 0 to 1 scale. Lower values suggest drier substrate; higher values suggest wetter substrate.
- **Substrate EC (substrateec):** Electrical conductivity of the root-zone solution, indicating nutrient concentration and salt load.
- **Substrate pH (substrateph):** The acidity or alkalinity of the substrate solution.
- **Canopy temperature (canopytempc):** The temperature of the crop canopy or leaf surface. It can differ from air temperature and may indicate crop stress.
- **Sensor density per 100 m2:** The number of sensors installed per 100 square metres of greenhouse area. Higher density means more granular monitoring.

Stress and risk measures

- **Pest pressure index (pestpressureindex):** A modeled score from 0 to 1 showing expected pest pressure, where 0 means no risk and 1 means extreme risk.
- **Disease risk index (diseaseriskindex):** A modeled score from 0 to 1 showing expected disease risk, where 0 means no risk and 1 means extreme risk.

- **Plant stress index (plantstressindex):** A composite 0 to 1 score summarizing overall crop stress, where 0 means healthy and 1 means critical stress.
- **Modeled index:** A calculated score built from underlying signals rather than a directly observed physical quantity. In this case, pest pressure, disease risk, and plant stress are modeled summary indicators.

Alerts and recommendations

- **Alert:** A warning generated when sensor conditions indicate a potentially important issue, such as high VPD, low moisture, high canopy temperature, or high pest pressure.
- **Alert type (alerttype):** The category of issue that triggered the warning, for example High VPD, Low Moisture, High Canopy Temp, or High Pest Pressure.
- **Alert flag (alertflag):** A 0/1 field showing whether an alert was active that day for that plot, where 1 means an alert occurred and 0 means none occurred.
- **Recommended action (recommendedaction):** The response suggested by the monitoring system, such as Increase Irrigation, Increase Venting, or Adjust Shade Venting.
- **Action taken (actiontaken):** A 0/1 field showing whether the recommended or related action was actually carried out.
- **Action delay days (actiondelaydays):** The number of days between the alert and the action. A value of 0 means same-day response.
- **Post-action stress delta 3d (postactionstressdelta3d):** The change in plant stress over the three days after the action window. Negative values usually mean stress improved; positive values mean stress worsened.
- **Response time:** A dashboard-friendly interpretation of actiondelaydays, showing how quickly the team reacted after an alert.
- **Same-day response:** A case where actiondelaydays = 0, meaning the alert and the intervention happened on the same day.

Management action terms

- **Input application:** A recorded management event in which a resource such as fertilizer, pesticide, water, labor, or energy is applied or used on a plot.
- **Input type (inputtype):** The category of resource used, including Fertilizer, Pesticide, Water, Labor, and Energy.
- **Quantity:** The amount of the input applied. Its unit depends on the input type, for example kilograms, litres, labour hours, or kilowatt-hours.
- **Unit cost (unitcost):** Cost per unit of the input, expressed in CAD-based units such as CAD/kg, CAD/L, CAD/hour, or CAD/kWh.
- **Total cost (totalcost):** The cost of one application event, typically calculated as quantity multiplied by unit cost.

- **Reason:** The immediate operational trigger for the application, such as Shock response, Growth phase, Pest response, Irrigation adjustment, or Routine.
- **Purpose:** The higher-level intent of the application, mainly Routine versus Response to alert.
- **Routine action / Routine management:** A regularly scheduled or background management activity that is part of normal operations, not necessarily triggered by a specific sensor alert.
- **Precision action:** A targeted intervention identified by $\text{isprecisionaction} = 1$. In this case, it is an application considered part of precision management, often linked to alert-driven or especially targeted decision making.
- **Response-to-alert action:** An intervention whose purpose is Response to alert, meaning it was made because the monitoring system or management process flagged a specific issue.
- **Targeted intervention:** A general term for a management action aimed at a specific problem, plot, or time rather than broad standard practice. In the case, this idea is closely related to precision actions and alert responses.
- **Shock response:** An application reason indicating a quick operational response to an acute issue or sudden change in conditions.
- **Growth phase:** An application reason indicating the input was timed to support a stage of crop development rather than to solve an urgent problem.
- **Pest response:** An application reason indicating the action was taken because of pest pressure or pest risk.
- **Irrigation adjustment:** An application reason indicating the action was taken to correct or fine-tune watering conditions.

Human observation terms

- **Scouting observation:** A human agronomist or scout's inspection record for a plot. These observations are less frequent than daily sensors and provide visual crop context.
- **Blossom count (blossomcount):** The number of blossoms per plant observed during scouting, often used as a leading indicator of yield potential.
- **Pest severity (pestseverity):** A visual 0 to 1 rating of pest damage or pressure, where 0 means none and 1 means severe.
- **Disease severity (diseaseseverity):** A visual 0 to 1 rating of disease symptoms, where 0 means none and 1 means severe.
- **Leaf color (leafcolor):** A categorical health signal such as Excellent, Good, Fair, or Poor based on observed foliage color.
- **Notes flag (notesflag):** A coded summary of issue severity from scouting notes, such as None, Minor, Moderate, or Severe.

Price and cost context

- **Market and input prices:** External benchmark prices for items such as fertilizer, pesticide, energy, and water, used to add cost context beyond individual applications.
- **Item:** The priced resource in the market-and-input-prices table, such as fertilizer, pesticide, energy, or water.
- **Price:** The observed unit price for the item on that date.
- **Benchmarking:** Comparing internal application costs or usage patterns against external market price context.

End-of-season outcome terms

- **Season summary:** The one-row-per-plot table of final production, quality, and financial outcomes for the season.
- **Season yield kg/m2 (seasonyieldkgm2):** Total production per square metre for the plot across the full season.
- **Marketable ratio (marketableratio):** The share of total production that is marketable, measured from 0 to 1, where 1 means all produce met market standards.
- **Season revenue CAD (seasonrevenuecad):** Total plot revenue for the season.
- **Total cost CAD (totalcostcad):** Total seasonal cost allocated to the plot, including inputs and fixed allocations.
- **Season profit CAD (seasonprofitcad):** Seasonal profit for the plot, generally revenue minus cost.
- **Season ROI (seasonroi):** Return on investment for the plot, defined in the case as $ROI = (\text{revenue} - \text{cost}) / \text{cost}$.
- **Precision benefit CAD (precisionbenefitcad):** Estimated dollar benefit attributed to precision management compared with routine management.
- **Marketable yield:** Not a separate field, but a useful dashboard concept combining yield and marketable ratio to show how much of the harvested output could actually be sold at premium quality.

Infrastructure terms

- **Structure type:** The greenhouse construction material or design, such as Glass or Polyfilm, used to compare infrastructure effects.
- **Heating system:** The greenhouse heating technology, such as Hydronic, Natural Gas, or Heat Pump.
- **Irrigation system:** The water delivery technology, such as Drip, EBB-Flow, or Fertigation.
- **Capacity plants (capacityplants):** Approximate maximum plant capacity of a greenhouse.

- **Greenhouse acres (greenhouseacres):** Farm-level greenhouse area in acres, useful for scale comparisons and normalization.

Data modeling and dashboard terms

- **Dimension:** A descriptive field used for grouping, filtering, slicing, or labeling data, such as farm, crop, greenhouse, alert type, or treatment.
- **Measure:** A numeric field used for calculation, aggregation, or comparison, such as cost, yield, stress score, or temperature.
- **Indicator / Flag:** A field coded as 0 or 1 to show whether a condition occurred, such as alertflag, actiontaken, or isprecisionaction.
- **Key:** A unique identifier used to distinguish records, such as farmid, greenhouseid, or plotid.
- **Foreign key:** A field used to connect one table to another, such as farmid in the greenhouse table or plotid in the daily readings table.
- **Join:** Linking tables using shared identifiers, such as joining plots to daily readings on plotid, or greenhouses to farms on farmid.
- **Roll-up:** Aggregating detailed data to a higher level, such as rolling plot results up to greenhouse or farm totals.
- **Normalization:** Adjusting values to a common scale, such as revenue per acre, yield per square metre, or cost per plant, so comparisons are fairer across different-sized units.
- **Plot-level vs greenhouse-level vs farm-level:** These labels show the grain of the data. Plot-level means one experimental microplot, greenhouse-level means one greenhouse unit, and farm-level means one whole operating site.
- **Grain:** The level of detail represented by one row in a table, such as one row per plot per day in daily sensor readings or one row per plot for the season in season summary.
- **Time series:** Data observed across dates, especially useful for line charts of daily stress, alerts, actions, or costs.
- **End-of-season outcome:** Final performance measures observed after the season ends, such as yield, profit, ROI, and precision benefit.
- **In-season performance:** Measures observed during the growing season, such as alerts, stress levels, action timing, and scouting scores.

Suggested interpretation notes

- **Precision actions vs routine actions:** Students should treat these as different management styles. Routine actions represent background operating practice, while precision actions represent more targeted, data-driven, or alert-linked interventions that may be evaluated for payoff.
- **Recommended action vs action taken:** These are not the same field. A recommendation is what the system suggested; action taken shows whether management actually acted.

- **Alert flag vs alert type:** alertflag only shows whether an alert occurred, while alert type describes which issue triggered it.
- **Reason vs purpose:** reason explains the immediate trigger in operational language, while purpose groups actions more broadly into routine work or alert response.
- **Plot as data entity vs plot as chart:** In this case, “plot” usually means an experimental microplot in the greenhouse, not a graph or chart. Students building dashboards should be careful with this double meaning.